

# Exact circuit construction of $Q^\dagger$ (rev. 4)

the inverse space-group symmetry-adaptation transform  
 $|L\rangle |a\rangle \mapsto |\mathbf{k}\rangle |I\rangle$ ,  $\mathbf{k} = (k_1, k_2, k_3)$  a clean 3D mesh index

Diamond,  $6 \times 6 \times 6$   $k$ -mesh ( $N_k = 216$ ,  $n_{\text{IP}} = 136$ ,  $|P| = 48$ )

Generated August 18, 2026 — costs computed on the actual bloqs

## Abstract

Rev. 4 settles the accounting after a careful pass over each stage. Included, all confirmed: (i) the Clifford-controlled phase-gradient rotation of arXiv:2007.07391 App. A ( $b-2$  Toffoli, implemented as **SignedCtrlAddIntoPhaseGrad** and substituted everywhere); (ii)  $P(g)$  is **charged**, not free; (iii)  $\mathbf{k}$  is output as a clean 3D mesh index built by the affine map  $\mathbf{k} = A_g \mathbf{k}_f$  (fractional/crystal coordinates, ready for the QFT); (iv) the ragged staircase is a  $\sqrt{\#\text{blocks}}$  binary tree; (v) the star member  $\kappa$  is a **compact, family-dependent** index (not the global op  $g$ ), so  $g$  comes from a small  $(f, \kappa) \rightarrow g$  lookup; (vi)  $(f, \kappa)$  is erased by **reading it back from  $\mathbf{k}$**  (a 216-entry lookup,  $\sim 20\times$  cheaper than a  $|P|$ -op argmin at this  $N_k$ ); (vii) the mesh uses the global largest-first ordering. **Deliberately excluded** (available but not counted): hard-coding size-1/size-2 orbits, and deduping “identical” families — the latter is not safe in general (same (star,  $|LG|$ , orbits) does not imply conjugate little groups). Result:  $Q^\dagger \approx \mathbf{11,150}$  Toffoli (exchange),  $\approx \mathbf{10,132}$  (direct).

## Contents

|          |                                                                               |          |
|----------|-------------------------------------------------------------------------------|----------|
| <b>1</b> | <b>Object, registers, conventions</b>                                         | <b>1</b> |
| <b>2</b> | <b>Primitive library</b>                                                      | <b>2</b> |
| <b>3</b> | <b><math>Q^\dagger</math> step by step</b>                                    | <b>2</b> |
| <b>4</b> | <b>Cost (computed; <math>6 \times 6 \times 6</math>, <math>b = 32</math>)</b> | <b>3</b> |

## 1 Object, registers, conventions

$Q : |\mathbf{k}\rangle |I\rangle \mapsto |i\rangle |a\rangle |c\rangle$ ,  $i = (f, \mu)$ ,  $a = (\kappa, \nu)$ ,  $c = \text{copy}$ .  $Q^\dagger$  is the inverse: fused irrep register  $|L\rangle = |(i, c)\rangle$  plus separate partner  $|a\rangle$ , out to  $|\mathbf{k}\rangle |I\rangle$ .

| register                       | holds                                                  | width                                 | note                      |
|--------------------------------|--------------------------------------------------------|---------------------------------------|---------------------------|
| L                              | fused $(i, c)$ , largest first                         | $\lceil \log_2 l_{\text{pad}} \rceil$ | pad pow-2 (X)/even (W)    |
| a                              | $(\kappa, \nu)$ , $\dim = m_i =  \text{star}_f  d_\mu$ | $\lceil \log_2 \max_i m_i \rceil = 5$ | Schur spectator           |
| $\mathbf{k} = (k_1, k_2, k_3)$ | momentum, 3D mesh index                                | $3 \times 3 = 9$                      | <b>clean 3D (for QFT)</b> |
| I                              | interpolation point                                    | 8                                     | orbit-major               |
| pg                             | phase gradient                                         | $b = 32$                              | returned clean            |

$\kappa$  is compact and family-dependent;  $g$  is not  $\kappa$ . Decoding  $a$  gives  $\kappa = a \text{ div } d_\mu \in \{0, \dots, |star_f| - 1\}$  — a compact index whose range depends on the family. Only  $|star_f|$  of the 48 ops are valid coset reps for  $f$ , so labelling  $\kappa$  by the global op  $g$  would over-pad  $\mathbf{a}$  (worst at high-symmetry families, e.g.  $\Gamma$  wastes 47/48). Instead the coset op  $g$  is obtained from a small  $(f, \kappa) \rightarrow g$  lookup ( $\sum_f |star_f| = 216$  rows). Output  $g$  is the *global* op index, so the geometry tables ( $A_g, \pi_g, L$ ) stay 48-addressed and  $N_k$ -independent. One such lookup serves both Step 2 and Step 3.

**k is a fractional mesh index, not Cartesian.**  $\mathbf{k}$  holds  $(n_1, n_2, n_3) \in \{0..5\}^3$  in the primitive reciprocal basis. This is required twice: the mesh is a regular integer grid closed under  $P$  (so  $A_g \cdot \mathbf{k} \bmod 6$  lands on a grid point by arithmetic, no table), and the QFT is a per-axis integer DFT over these indices. In this basis  $A_g$  is a fixed small-integer matrix (entries in  $\{-1, 0, 1\}$  for cubic), so  $A_g \mathbf{k}_f$  is a few controlled adds, not a lookup.

## 2 Primitive library

**P-A. Interferometer (Givens) unitary synthesis.**  $n \times n$  unitary as a Clements/Reck mesh (Optica **3**,1460 (2016); PRL **73**,58 (1994)), block-aware QROAM-loaded angles (arXiv:2409.11748). Per-layer cost =  $\frac{D}{2\Lambda} + 2\Lambda b$  ( $D = n_{\text{rows}}$ ,  $\Lambda$  select-swap optimum), incl. the two App. A rotations; overall  $\sim n^{3/2} \sqrt{b}$  (verified exponent  $\rightarrow 1.50$ ). Bloqs `BlockInterferometerPhaseLayerQROAM + ...FinalPhasesQROAM`.

**P-B. QROAM (select-swap, clean).**  $d$  words  $\times w$  bits:  $\approx \lceil d/\lambda \rceil + \lambda w$ ,  $\lambda$  optimal (arXiv:1812.00954, 1902.02134). All  $\lambda$  here brute-forced = minimum.

**P-C. Controlled phase-gradient rotation (arXiv:2007.07391 App. A).** Controlled rotation to  $b$  bits costs  $b - 2$  Toffoli: the control drives add $\leftrightarrow$ subtract `cnots` (two's complement) around one *uncontrolled* addition; the  $-2$  is the MSB- $|+\rangle$  discard + carry $\rightarrow$ controlled-phase. Bloq `SignedCtrlAddIntoPhaseGate` = 30 at  $b = 32$  vs 60 for the naive controlled adder. Implemented and substituted at every controlled-rotation site.

**P-D.  $\sqrt{\#\text{blocks}}$  staircase (binary tree).** Ragged $\rightarrow$ padded index map in 2 levels: with pow-2 buddy alignment the leaf index is a free bit-slice (no compares), a QROAM loads per-leaf offsets, and  $\sim \sqrt{m}$  compares/adds finish it.  $\approx 2\sqrt{m}(w - 1) + \text{QROAM}(\sqrt{m})$ ;  $m = 49 \Rightarrow 161$  (vs 1056 linear).

**P-E. Multiplexed affine map  $A_g$ .** SELECT over the 48 fixed integer matrices, applied in place to a coordinate register (few controlled adds; signed-perm/Clifford in the cubic Cartesian basis). Used for  $\mathbf{k} = A_g \mathbf{k}_f$  and, optionally, the grid  $\pi_g$ .

**P-F. Comparator / controlled add / lookup permutation.** `LessThanConstant`, `AddK`, and in-place table permutations (forward + inverse lookup).

## 3 $Q^\dagger$ step by step

**Input.**  $|L\rangle |a\rangle |0\rangle_{\mathbf{k}} |0\rangle_I |\Gamma\rangle_{\text{pg}}$ .

**Step 1 — Unpack  $L \rightarrow (f, \mu, c)$  (P-D, P-B, P-F)**

$\sqrt{m}$ -tree staircase splits the fused index into (block, copy); a 49-entry QROAM maps  $\text{block} \rightarrow (f, \mu)$ . Because  $i \leftrightarrow (f, \mu)$  is a bijection, the ordinal is erased in place (forward + inverse lookup), not merely copied.

The input labels  $(f, \mu, \kappa, \nu, c)$  split into two parallel legs:  $(f, \kappa) \rightarrow \mathbf{k}$  (Step 2) and  $(\mu, \nu, c) \rightarrow I$  (Step 3), sharing  $f$  and  $\kappa$  (erased last, Step 4).

**Step 2 — k-leg: build  $\mathbf{k} = A_g \mathbf{k}_f$  (P-B, P-E)**

Decode  $\kappa$  from  $a$  (compact);  $g = (f, \kappa) \rightarrow g$  lookup. Load  $\mathbf{k}_f$  (16-entry) and apply  $\mathbf{k} = A_g \mathbf{k}_f \bmod 6$  (P-E) — a clean 3D mesh index, ready for the QFT.  $f$  and  $\kappa$  are *kept* (needed by Step 3).

**Step 3 — I-leg: reindex  $\rightarrow$  mesh  $\rightarrow P(g)$ , then fold-back phase (P-B, P-A, P-E)**

This is where  $I$  is produced, from the fiber labels  $(\mu, \nu, c)$ :

1. **reindex** irrep-major  $(\mu, \nu, c) \rightarrow$  orbit-major slot (fused permutation over the global largest-first index, family  $f$ );
2. **block unitary**  $\bigoplus_o U_o^\dagger$  (fused mesh, all 730 orbits, 48 layers, family  $f$  selected)  $\Rightarrow$  the anchor fiber  $I'$  (orbit-major, so  $J = I'$ );
3. **grid permutation**  $P(g)$ :  $\pi_g$  carries  $I' \rightarrow I$  (the actual star-member interpolation index);  $\pi_g$  is a  $48 \times 136$  QROAM on  $(g, I')$  (or affine  $A_g$  on coordinates);
4. **fold-back phase**  $e^{-i\mathbf{k} \cdot L_{I'}(g)}$ :  $L$  from the  $48 \times 136$  table on  $(g, I')$  (or the affine mod-cell reduction),  $\mathbf{k} \cdot L$  into  $\text{pg}$  by App. A rotations.

**Step 4 — erase  $(f, \kappa)$ , uncompute scratch (P-F)**

Now that both  $\mathbf{k}$  and  $I$  exist and  $(f, \kappa)$  are no longer needed: since  $(f, \kappa) \leftrightarrow \mathbf{k}$  is a bijection, read  $(f, \kappa)$  back from  $\mathbf{k}$  (a 216-entry lookup) and XOR-cancel. (No  $|P|$ -op argmin — that canonicalization is  $\sim 20\times$  more expensive at  $N_k = 216$ .) Uncompute remaining scratch.

**Output.**  $|\mathbf{k}\rangle |I\rangle |\Gamma\rangle_{\text{pg}}$ , registers separate,  $\mathbf{k}$  a clean 3-vector.

**4 Cost (computed;  $6 \times 6 \times 6$ ,  $b = 32$ )**

Measure  $n_{\text{ccz}} + n_t/4$ , all  $\lambda$  verified optimal.

| stage                                                           | Toffoli                            | note  |
|-----------------------------------------------------------------|------------------------------------|-------|
| Step 3 — orbit mesh $\bigoplus_o U_o^\dagger$ (all 730, App. A) | 8,670                              | exact |
| Step 3 — reindex (fused permutation)                            | 420                                | exact |
| Step 3 — $P(g)$ grid permutation $48 \times 136$                | 612                                | exact |
| Step 3 — fold-back $L_I(g)$ $48 \times 136$                     | 928                                | exact |
| Step 3 — Bloch phase $3(b-2)$                                   | 90                                 | exact |
| Step 2 — $\mathbf{k} = A_g \mathbf{k}_f$                        | $\sim 68$                          | est   |
| Step 2 — $(f, \kappa) \rightarrow g$ lookup (used in Steps 2,3) | 93                                 | exact |
| Step 1 — unpack $L$ ( $\sqrt{m}$ -tree)                         | 161                                | exact |
| Step 4 — erase $(f, \kappa)$ (read from $\mathbf{k}$ )          | 108                                | exact |
| <b>Q<sup>†</sup> total (exchange)</b>                           | <b><math>\approx 11,150</math></b> |       |
| <b>Q<sup>†</sup> total (direct: fold-back phase cancels)</b>    | <b><math>\approx 10,132</math></b> |       |

**Trajectory.** 12,214 (rev 1,  $P(g)$  wrongly free)  $\rightarrow \sim 13,900$  (rev 2,  $P(g)$  charged)  $\rightarrow \sim 10,900$  (rev 3, App. A)  $\rightarrow$  **11,150** (rev 4: + compact- $\kappa$   $(f, \kappa) \rightarrow g$  lookup, read-from- $\mathbf{k}$  erase, honest reindex/orderings). The small rise over rev 3 is the erase and  $g$ -lookup that rev 3 had omitted.

**Excluded levers (available, not counted).** (1) **Hard-code size-1 (identity, free) and size-2 (Hadamard) orbits** — proven correct, saves  $\sim 752$  off the mesh, omitted here to keep the mesh a single uniform synthesis. (2) **Family dedup** — *not safe in general*: same (star,  $|LG|$ , orbit sizes) does not imply conjugate little groups ( $O_h$  has non-conjugate  $\sigma_h$  vs  $\sigma_d$ ), so “identical-looking” families can have different  $U_o$  for size- $\geq 3$  orbits; only the size- $\leq 2$  families are provably identical. (3) **Affine  $P(g)$**  — would cut the 612+928 tables toward a few hundred but is coordinate-dependent; the table route above is the exactly-computed number. With (1)+(3) the total would drop toward  $\sim 8.5$ –9k, but those are not counted until verified.

**$N_k$ -independence.** No stage carries an  $N_k$  factor: the mesh/reindex are addressed by the fixed  $16 \times 136$  family/orbit structure,  $P(g)$  and  $A_g$  by the 48 ops, and  $\mathbf{k}$  is computed by arithmetic. The 216-entry  $(f, \kappa) \rightarrow g$  and read-from- $\mathbf{k}$  lookups scale as  $\sqrt{N_k}$  (select-swap), so they stay cheap far beyond any practical mesh.